UNIT 3 · TOPIC 7(B)

Harvesters and Combines — Loss Calculations

Syllabus topic: Calculation of capacity, efficiency and losses in threshers, harvesters and chaff cutters

Converting a grain count to a loss

1. Count the grains per m² behind the machine and subtract the pre-harvest count (Topic 4a).
2. Multiply by the single-grain mass (1000-grain weight ÷ 1000) → g/m².
3. Multiply g/m² by 10 → kg/ha.
4. Divide by the yield → % loss.

Worked example. 60 wheat grains/m² (after subtracting pre-harvest), 1000-grain weight 40 g → 60 × 0.04 = 2.4 g/m² = **24 kg/ha**; at 4 t/ha yield → total machine loss = **0.6%**.

- Acceptable total combine loss: **≤ ~3%**.

Practice MCQs

Q83. 60 grains/m² of wheat (1000-grain wt 40 g) equals a loss of:

(a) 2.4 kg/ha (b) 12 kg/ha
(c) 24 kg/ha (d) 60 kg/ha

Q84. If crop yield is 4 t/ha, a 24 kg/ha machine loss is:

(a) 0.06% (b) 0.6%
(c) 6% (d) 2.4%

Q85. Acceptable TOTAL combine loss is commonly taken as up to about:

(a) 0.5% (b) 3%
(c) 10% (d) 15%

Q104. Grain loss in g/m² is converted to kg/ha by multiplying by:

(a) 1 (b) 10
(c) 100 (d) 1000

Answers

Q83 — (c) 24 kg/ha — *60 × 0.040 g = 2.4 g/m²; ×10 → 24 kg/ha.*

Q84 — (b) 0.6% — *24/4000 = 0.6%.*

Q85 — (b) 3%

Q104 — (b) 10 — *g/m² × 10 = kg/ha (1 ha = 10,000 m²; 1 kg = 1000 g).*

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.